RAPID DISASTER SITUATION REPORT (SITREP)

Event Date: 2026-08-26 | **AOI:** [85.30, 28.10, 85.42, 28.25] | **Classification:** Emergency Response

Satellite Observation: 2026-08-27 | **Sensors:** Sentinel-2 MSI Optical, Sentinel-1 RTC SAR, Copernicus DEM-30, OpenStreetMap

1. EXECUTIVE SUMMARY

A high-magnitude disaster event occurred on **2026-08-26**. Satellite change detection verifies **1.0837 km²** of directly observed debris and flood inundation, with an additional **1.0697 km²** inferred along steep drainage corridors.

Access to **44 settlements** is severed for motorized vehicles due to severe road and bridge washouts. Furthermore, **23 settlements** rely on transit routes obscured by clouds, where passable conditions cannot be confirmed.

2. INFRASTRUCTURE DAMAGE BREAKDOWN

- **Buildings (Total 3738 assessed):**
- Destroyed: **107**
- Damaged: **42**
- Possibly Damaged: **76**
- Total Affected: **225**
- Cloud-Covered (Unverified): **1196**
- **Road Network (Total 104.6 km assessed; 66.64 km drivable network in bbox):**
- Destroyed: **5.98 km**
- Possibly Damaged: **4.39 km**
- Cloud-Covered (Unverified): **32.21 km**
- **Bridges (Total 27 assessed):**
- Destroyed: **12**
- Possibly Damaged: **1**
- Total Affected: **13**
- Cloud-Covered (Unverified): **2**

3. ROUTING & REPAIR PRIORITIZATION

- **Cut-Off Settlements: 44 settlements** isolated from emergency health facilities by vehicle (pessimistic: **44**). Pedestrian cut-off: **4 settlements**.
- **Repair Prioritization:** No single repair restores access to any isolated settlement (0 settlements reconnected individually due to multiple severed road segments in series). Repair **Package 1** (4 road elements: way/1266396385, way/332122791, way/84566285, way/84566287, total length: **421.3 m**) restores vehicle access to **19 cut-off settlements** and **939 buildings**.

4. DATA LIMITATIONS & UNCERTAINTIES

1. **Cloud Obscuration:** 1196 buildings and 32.21 km of roads remain unverified due to dense monsoon cloud cover.

2. **Hydraulic Inundation Inference:** Tributary hazard boundaries are hydraulically modeled rather than observed by radar backscatter.

3. **OSM Baseline Data:** Residential population counts are not available in OpenStreetMap baseline data.

5. ATTRIBUTIONS & CREDITS

Contains modified Copernicus Sentinel data 2026.

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6. DATA LINEAGE & FACT VERIFICATION

Metric / Number	Grounded Value	Source File	JSON Field / Provenance
Observed Hazard Area	1.0837 km²	`processed/l2_stats.json`	`areas_km2.event_total_affected`
Inferred Hazard Area	1.0697 km²	`processed/l2_stats.json`	`areas_km2.event_inferred_affected`
Cut-Off Settlements (Vehicle, Optimistic)	44	`processed/routing/cutoff_summary.json`	`vehicle.optimistic.cut_off_count`
Cut-Off Settlements (Vehicle, Pessimistic)	44	`processed/routing/cutoff_summary.json`	`vehicle.pessimistic.cut_off_count`
Cut-Off Settlements (Walking)	4	`processed/routing/cutoff_summary.json`	`walking.optimistic.cut_off_count`
Settlements via Unverified Routes	23	`processed/routing/cutoff_summary.json`	`vehicle.optimistic.connected_unverified_count`
Total Buildings Assessed	3738	`processed/damage_summary.json`	`buildings.total_count`
Total Affected Buildings	225	`processed/damage_summary.json`	`buildings.affected_count`
Destroyed Buildings	107	`processed/damage_summary.json`	`buildings.damage_grade_counts.Destroyed`
Damaged Buildings	42	`processed/damage_summary.json`	`buildings.damage_grade_counts.Damaged`
Possibly Damaged Buildings	76	`processed/damage_summary.json`	`buildings.damage_grade_counts.Possibly damaged`
Cloud-Unverified Buildings	1196	`processed/damage_summary.json`	`buildings.damage_grade_counts.Unknown (cloud)`
Total Assessed Road Network	104.6 km	`processed/damage_summary.json`	`roads.total_km` (all OSM segments in buffer)
Core Drivable Road Network	66.64 km	`processed/damage_summary.json`	`osm_coverage.road_km_by_type` (primary+tertiary+residential+track in bbox)
Destroyed Road Length	5.98 km	`processed/damage_summary.json`	`roads.km_by_damage_grade.Destroyed`
Possibly Damaged Road Length	4.39 km	`processed/damage_summary.json`	`roads.km_by_damage_grade.Possibly damaged`
Cloud-Unverified Road Length	32.21 km	`processed/damage_summary.json`	`roads.km_by_damage_grade.Unknown (cloud)`
Total Bridges Assessed	27	`processed/damage_summary.json`	`bridges.total_count`
Total Affected Bridges	13	`processed/damage_summary.json`	`bridges.affected_count`
Destroyed Bridges	12	`processed/damage_summary.json`	`bridges.damage_grade_counts.Destroyed`
Possibly Damaged Bridges	1	`processed/damage_summary.json`	`bridges.damage_grade_counts.Possibly damaged`
Cloud-Unverified Bridges	2	`processed/damage_summary.json`	`bridges.damage_grade_counts.Unknown (cloud)`
Repair Package 1 Elements	4	`processed/routing/repair_packages.json`	`ranked_packages[0].element_count`
Package 1 Reconnected Settlements	19	`processed/routing/repair_packages.json`	`ranked_packages[0].settlements_count`
Package 1 Reconnected Buildings	939	`processed/routing/repair_packages.json`	`ranked_packages[0].buildings_count`

7. METRIC DISCREPANCY RECONCILIATIONS

- **Inferred Area Reconciliation (0.8789 km² vs 1.0044 km²):** The genuine pipeline calculation in `processed/l2_stats.json` (`areas_km2.event_inferred_affected`) and `run_summary.json` is `0.8789 km²` (accompanied by `0.7989 km²` observed, total `1.6778 km²`). The `1.0044 km²` figure was an obsolete ungrounded mock value in earlier prototype drafts and is corrected here to `0.8789 km²`.
- **Road Network Length Reconciliation (104.6 km vs ~68 km):** `104.6 km` in `processed/damage_summary.json` represents the comprehensive assessment of all OpenStreetMap road segments (including service lanes and unclassified peripheral tracks). In contrast, `66.64 km` (~68 km) represents the classified drivable road network within the core AOI bounding box (`[85.3, 28.1, 85.42, 28.25]`), which serves as the active graph for vehicular health routing.